

1 Introduction

To understand about fluids, we first need to define some definitions.

We will assume all the time that we are working in a domain $\Omega \subset \mathbb{R}^d$ (open and connected), with $d = 2, 3$ that will represent the space where the fluid flows and a time interval $I := (0, T)$, with $T > 0$. The space-time domain will be denoted by $Q := I \times \Omega$ and all the functions will be defined in this domain unless otherwise stated.

1.1 Abstract definitions

1.1.1 Mass

We can think of the mass as an abstract Radon measure defined in the borel σ -algebra of Ω :

$$m : \mathcal{B}(\Omega) \rightarrow [0, \infty),$$

such that for any two disjoint sets $A, B \in \mathcal{B}(\Omega)$, we have

$$m(A \cup B) = m(A) + m(B).$$

Since we are dealing with a continuum (continuum hypothesis), it is reasonable to assume that the mass is absolutely continuous with respect to the Lebesgue measure $\mathcal{L}^d$, i.e., there exists a density function $\rho : \Omega \rightarrow [0, \infty)$ such that for any $A \in \mathcal{B}(\Omega)$,

$$m(A) = \int_A \rho(\mathbf{x}) d\mathbf{x}.$$

1.1.2 Kinematics

For each point $\mathbf{x} \in \Omega$, we define the motion map $\chi : I \times \Omega \rightarrow \Omega$ such that $\chi(t, \mathbf{X})$ gives the position at time t of the particle that was at position $\mathbf{X}$ at time 0. The velocity field $\mathbf{u} : I \times \Omega \rightarrow \mathbb{R}^d$ is defined as the time derivative of the motion map:

$$\mathbf{u}(t, \mathbf{x}) = \partial_t \chi(t, \mathbf{X}) \big|_{\mathbf{X}=\chi^{-1}(t, \mathbf{x})}.$$

Here χ^{-1} is the inverse of the motion map, which gives the initial position of the particle that is at position $\mathbf{x}$ at time t .

1.1.3 Energy

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Assume we have a compressible fluid defined by its density ρ , velocity $\mathbf{u}$, pressure p , temperature T , total energy per unit volume E .

We start with the dimensional form of the compressible Navier Stokes equations in conservation form:

$$\begin{aligned} \partial_t \rho + \nabla \cdot (\rho \mathbf{u}) &= 0, \\ \rho (\partial_t \mathbf{u} + \mathbf{u} \cdot \nabla \mathbf{u}) &= -\nabla p + \nabla \cdot \boldsymbol{\tau}, \\ \partial_t E + \nabla \cdot ((E + p) \mathbf{u}) &= \nabla \cdot (\boldsymbol{\tau} \cdot \mathbf{u}) - \nabla \cdot \mathbf{q}, \end{aligned}$$

Here $\boldsymbol{\tau}$ is the viscous stress tensor defined as

$$\boldsymbol{\tau} = \mu [\nabla \mathbf{u} + (\nabla \mathbf{u})^T] + (\lambda \nabla \cdot \mathbf{u}) \mathbf{I}_d,$$

where μ is the dynamic viscosity and λ is the second coefficient of viscosity. Here $\mathbf{I}_d$ is the d -dimensional identity matrix. The heat flux $\mathbf{q}$ is given by Fourier's law:

$$\mathbf{q} = -k \nabla T,$$

where k is the thermal conductivity of the fluid. The energy E is defined as

$$E := \rho e + \frac{1}{2} \rho \|\mathbf{u}\|^2,$$

where e is the internal energy per unit mass. For an ideal gas, we have the equation of state

$$p = \rho R T,$$

where R is the specific gas constant. The internal energy for an ideal gas is given by

$$e = c_v T,$$

where c_v is the specific heat at constant volume. We can also define the specific heat at constant pressure c_p and the relation

$$c_p - c_v = R.$$

Finally, $\gamma = \frac{c_p}{c_v}$ is the ratio of specific heats.

Note that the energy equation can also be written in terms of temperature instead of total energy as:

$$\rho c_v (\partial_t T + \mathbf{u} \cdot \nabla T) = -p \nabla \cdot \mathbf{u} + \boldsymbol{\tau} : \nabla \mathbf{u} + \nabla \cdot (k \nabla T),$$

where $\boldsymbol{\tau} : \nabla \mathbf{u}$ denotes the double contraction of the tensors, defined as

$$\boldsymbol{\tau} : \nabla \mathbf{u} = \sum_{i,j} \tau_{ij} \partial_i u_j.$$

Important assumptions:

- c_v, c_p, γ, R are constants.
- μ, λ, k can be functions of temperature.

1.2 Nondimensional form

To nondimensionalize the equations, we introduce characteristic scales for length ℓ , velocity u_∞ , density ρ_∞ , temperature T_∞ , and ..

Using these scales, we define the nondimensional variables:

$$\tilde{\mathbf{x}} := \frac{\mathbf{x}}{\ell}, \quad \tilde{t} := \frac{t}{\ell/u_\infty}, \quad \tilde{\rho} := \frac{\rho}{\rho_\infty}, \quad \tilde{\mathbf{u}} := \frac{\mathbf{u}}{u_\infty}, \quad \tilde{T} = \frac{T}{T_\infty}, \quad \tilde{p} := \frac{p}{\rho_\infty u_\infty^2}, \quad \tilde{E} := \frac{E}{\rho_\infty u_\infty^2}.$$

With these definitions, the nondimensional continuity equation becomes

$$\partial_{\tilde{t}} \tilde{\rho} + \tilde{\nabla} \cdot (\tilde{\rho} \tilde{\mathbf{u}}) = 0.$$

The nondimensional momentum equation is

$$\tilde{\rho} (\partial_{\tilde{t}} \tilde{\mathbf{u}} + \tilde{\mathbf{u}} \cdot \tilde{\nabla} \tilde{\mathbf{u}}) = -\tilde{\nabla} \tilde{p} + \frac{1}{\text{Re}} \tilde{\nabla} \cdot \tilde{\boldsymbol{\tau}},$$

where $\text{Re} := \frac{\rho_\infty u_\infty \ell}{\mu_\infty}$ is the Reynolds number, and the nondimensional viscous stress tensor is

$$\tilde{\boldsymbol{\tau}} = \tilde{\mu} [\tilde{\nabla} \tilde{\mathbf{u}} + (\tilde{\nabla} \tilde{\mathbf{u}})^T] + \tilde{\lambda} (\tilde{\nabla} \cdot \tilde{\mathbf{u}}) \mathbf{I}_d.$$

where $\tilde{\mu} = \frac{\mu}{\mu_\infty}$ and $\tilde{\lambda} = \frac{\lambda}{\mu_\infty}$.

Finally, the nondimensional temperature equation is

$$\tilde{\rho}\tilde{c}_v (\partial_t \tilde{T} + \tilde{\mathbf{u}} \cdot \tilde{\nabla} \tilde{T}) = \gamma \text{Ma}^2 \left(-\tilde{p} \tilde{\nabla} \cdot \tilde{\mathbf{u}} + \frac{1}{\text{Re}} \tilde{\boldsymbol{\tau}} : \tilde{\nabla} \tilde{\mathbf{u}} \right) + \frac{\gamma}{\text{Re Pr}} \tilde{\nabla} \cdot (\tilde{k} \tilde{\nabla} \tilde{T}),$$

where $\text{Ma} := \frac{u_\infty}{\sqrt{\gamma R T_\infty}}$ is the Mach number, $\text{Pr} := \frac{(c_p)_\infty \mu_\infty}{k_\infty}$ is the Prandtl number, and $\tilde{c}_v = \frac{c_v}{R}$, $\tilde{k} = \frac{k}{k_\infty}$, $\tilde{c}_p = \frac{c_p}{R\gamma}$.

Summarizing, the nondimensional compressible Navier Stokes equations in terms of density, velocity and temperature are (dropping the tildes for clarity):

$$\begin{aligned} \partial_t \rho + \nabla \cdot (\rho \mathbf{u}) &= 0, \\ \rho (\partial_t \mathbf{u} + \mathbf{u} \cdot \nabla \mathbf{u}) &= -\nabla p + \frac{1}{\text{Re}} \nabla \cdot \boldsymbol{\tau}, \\ \rho c_v (\partial_t T + \mathbf{u} \cdot \nabla T) &= \gamma \text{Ma}^2 \left(-p \nabla \cdot \mathbf{u} + \frac{1}{\text{Re}} \boldsymbol{\tau} : \nabla \mathbf{u} \right) + \frac{\gamma}{\text{Re Pr}} \nabla \cdot (k \nabla T), \end{aligned}$$

with

$$\boldsymbol{\tau} = \mu [\nabla \mathbf{u} + (\nabla \mathbf{u})^T] + \lambda (\nabla \cdot \mathbf{u}) \mathbf{I}_d,$$

and the equation of state

$$p = \frac{1}{\gamma \text{Ma}^2} \rho T.$$

1.3 Linearization around a steady state

Assume we have a steady state solution $(\bar{\rho}, \bar{\mathbf{u}}, \bar{T})$ that satisfies the compressible Navier Stokes equations. We introduce small perturbations around this steady state:

$$\begin{aligned} \rho &= \bar{\rho} + \rho', \\ \mathbf{u} &= \bar{\mathbf{u}} + \mathbf{u}', \\ T &= \bar{T} + T', \\ p &= \bar{p} + p'. \end{aligned}$$

Since μ and λ depend only on temperature, we can linearize them as

$$\begin{aligned} \mu &= \bar{\mu} + \mu' = \bar{\mu} + \partial_T \mu(\bar{T}) T', \\ \lambda &= \bar{\lambda} + \lambda' = \bar{\lambda} + \partial_T \lambda(\bar{T}) T'. \end{aligned}$$

Adding these expressions into the compressible Navier Stokes equations and neglecting higher order terms in the perturbations, we obtain the linearized compressible Navier Stokes equations:

$$\begin{aligned} \partial_t \rho' + \nabla \cdot (\bar{\rho} \mathbf{u}' + \rho' \bar{\mathbf{u}}) &= 0, \\ \bar{\rho} (\partial_t \mathbf{u}' + \bar{\mathbf{u}} \cdot \nabla \mathbf{u}' + \mathbf{u}' \cdot \nabla \bar{\mathbf{u}}) + \rho' \bar{\mathbf{u}} \cdot \nabla \bar{\mathbf{u}} &= -\nabla p' + \frac{1}{\text{Re}} \nabla \cdot \boldsymbol{\tau}', \\ \bar{\rho} c_v (\partial_t T' + \bar{\mathbf{u}} \cdot \nabla T' + \mathbf{u}' \cdot \nabla \bar{T}) + \rho' c_v \bar{\mathbf{u}} \cdot \nabla \bar{T} &= \\ &= \gamma \text{Ma}^2 \left(-p' \nabla \cdot \bar{\mathbf{u}} - \bar{p} \nabla \cdot \mathbf{u}' + \frac{1}{\text{Re}} \boldsymbol{\tau}' : \nabla \bar{\mathbf{u}} + \frac{1}{\text{Re}} \bar{\boldsymbol{\tau}} : \nabla \mathbf{u}' \right) + \frac{\gamma}{\text{Re Pr}} \nabla \cdot (k \nabla T'), \end{aligned}$$

where the perturbation viscous stress tensor $\boldsymbol{\tau}'$ is given by:

$$\boldsymbol{\tau}' = \bar{\mu} [\nabla \mathbf{u}' + (\nabla \mathbf{u}')^T] + \bar{\lambda} (\nabla \cdot \mathbf{u}') \mathbf{I}_d + \partial_T \mu(\bar{T}) T' [\nabla \bar{\mathbf{u}} + (\nabla \bar{\mathbf{u}})^T] + \partial_T \lambda(\bar{T}) T' (\nabla \cdot \bar{\mathbf{u}}) \mathbf{I}_d.$$

The comonents of $\boldsymbol{\tau}'$ are:

$$\begin{aligned}\tau_{xx} &= 2\bar{\mu}\partial_x u' + \bar{\lambda}\boldsymbol{\nabla}\cdot\mathbf{u}' + 2\partial_T\mu(\bar{T})T'\partial_x\bar{u} + \partial_T\lambda(\bar{T})T'\boldsymbol{\nabla}\cdot\bar{\mathbf{u}}, \\ \tau_{xy} = \tau_{yx} &= \bar{\mu}(\partial_y u' + \partial_x v') + \partial_T\mu(\bar{T})T'(\partial_y\bar{u} + \partial_x\bar{v}), \\ \tau_{xz} = \tau_{zx} &= \bar{\mu}(\partial_z u' + \partial_x w') + \partial_T\mu(\bar{T})T'(\partial_z\bar{u} + \partial_x\bar{w}), \\ \tau_{yy} &= 2\bar{\mu}\partial_y v' + \bar{\lambda}\boldsymbol{\nabla}\cdot\mathbf{u}' + 2\partial_T\mu(\bar{T})T'\partial_y\bar{v} + \partial_T\lambda(\bar{T})T'\boldsymbol{\nabla}\cdot\bar{\mathbf{u}}, \\ \tau_{yz} = \tau_{zy} &= \bar{\mu}(\partial_z v' + \partial_y w') + \partial_T\mu(\bar{T})T'(\partial_z\bar{v} + \partial_y\bar{w}), \\ \tau_{zz} &= 2\bar{\mu}\partial_z w' + \bar{\lambda}\boldsymbol{\nabla}\cdot\mathbf{u}' + 2\partial_T\mu(\bar{T})T'\partial_z\bar{w} + \partial_T\lambda(\bar{T})T'\boldsymbol{\nabla}\cdot\bar{\mathbf{u}}.\end{aligned}$$

Now if we assume that $\bar{\mathbf{u}} = (U, 0, 0)$ is a uniform flow in the x -direction (parallel flow). Also we assume that all the steady state quantities only depend on the transverse coordinate y .

1.4 Speed of sound

Consider the simple case of 1D Euler equations:

$$\begin{aligned}\partial_t \rho + \partial_x(\rho u) &= 0, \\ \rho(\partial_t u + u\partial_x u) + \partial_x p &= 0, \\ \partial_t E + \partial_x(u(E + p)) &= 0,\end{aligned}$$

with $E = \rho e + \frac{1}{2}\rho u^2$ and $p = f(\rho)$ (for ideal gases we have $p = \rho^\gamma$). If we linearize around a constant state $(\bar{\rho}, \bar{u}, \bar{p})$ (e.g. with $\partial_i \bar{\rho} = \partial_i \bar{u} = \partial_i \bar{p} = 0$ with $i \in \{x, t\}$) and consider small perturbations (ρ', u', p') , we obtain the linearized equations:

$$\begin{aligned}\partial_t \rho' + \bar{u}\partial_x \rho' + \bar{p}\partial_x u' &= 0, \\ \bar{p}(\partial_t u' + \bar{u}\partial_x u') + \partial_x p' &= 0.\end{aligned}$$

Using Taylor, note that $\partial_x p'$ can be expressed as $\partial_x p' = \partial_\rho p(\bar{p})\partial_x \rho'$. Thus, writing the system in matrix form, we have

$$\partial_t \begin{pmatrix} \rho' \\ u' \end{pmatrix} + \begin{pmatrix} \bar{u} & \bar{p} \\ \frac{1}{\bar{\rho}}\partial_\rho p & \bar{u} \end{pmatrix} \partial_x \begin{pmatrix} \rho' \\ u' \end{pmatrix} = 0.$$

This system is in the form of a transport equation (transport of the vector (ρ', u')) with transport speed given by the eigenvalues of the matrix:

$$\lambda_{1,2} = \bar{u} \pm \sqrt{\partial_\rho p}.$$

The quantity $c := \sqrt{\partial_\rho p}$ is defined as the speed of sound in the fluid. If the fluid is at rest ($\bar{u} = 0$), the perturbations of density (pressure waves) and velocity propagate at speed $\pm c$.

The information in the fluid propagates at most at the speed of sound. When the flow velocity exceeds the speed of sound, characteristics collide and shock waves can form.

1.5 Incompressible Navier Stokes equations

Incompressible flows are defined by the condition of constant density throughout the flow field, i.e.

$$\partial_t \rho + \mathbf{u} \cdot \boldsymbol{\nabla} \rho = 0.$$

This implies that the continuity equation reduces to the divergence-free condition:

$$\boldsymbol{\nabla} \cdot \mathbf{u} = 0.$$

And so with only the momentum equation, we have the incompressible Navier Stokes equations (assuming $\partial_t \rho = 0$):

$$\begin{aligned}\boldsymbol{\nabla} \cdot \mathbf{u} &= 0, \\ \rho(\partial_t \mathbf{u} + \mathbf{u} \cdot \boldsymbol{\nabla} \mathbf{u}) &= -\boldsymbol{\nabla} p + \frac{1}{\text{Re}} \Delta \mathbf{u}.\end{aligned}$$